

3.5 - Chain Rule

Goals: Compute the derivative of compositions of functions

Before Class Videos: 2.5.1, 2.5.2

Example 1: Consider $\sin(x^2)$. This is an example of a composition $f(g(x))$ or alternatively $f \circ g$.

Example 2: $f(x) = x \sin x$, $g(x) = \cos x + 1$. Write out:

- $f(g(x)) =$

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Theorem (Chain Rule):

If g differentiable at x and f differentiable at $g(x)$ then

$$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x).$$

Example 3: $\frac{d}{dx} \sin(x^2 + 1)$

Example 4: $\frac{d}{dx} \sin^{30}(x)$

Example 5: $\frac{d}{dx} \cos(3e^x)$

Example 6: $\frac{d}{dx} \frac{1}{2x+1}$

Example 7: $\frac{d}{dx} \cos^2(x^3)$

Example 8: $\frac{d}{dx} \tan(x^2 e^x).$

Leibniz Notation: let $u = g(x)$,

$$\frac{d}{dx}[f(g(x))] = \frac{df}{du} \frac{du}{dx}.$$

Example 9: Car uses $6L/100km$. Cost is $1.5\$/L$. What is the rate of change in cost per $100km$?

Example 10: Gravel is dumped from a conveyor to form pile with

$$V(h) = \frac{\pi}{12}h^3.$$

At $5m$ tall, the pile is growing by 0.1 m/s. How fast is volume growing?

Example 11: Contrast graphically the slope of $f(x)$ and $f(2x)$ for an arbitrary function $f(x)$.

Example 12: Derive the Quotient Rule.

Time Dependent Intro to Implicit Differentiation:

Example 13: Explore $x^2 + y^2 = 1$. Where does it have vertical and horizontal tangents?

Big Idea: Treat $y = y(x)$ when x is independent and y is dependent.

Example 14: Now explore

$$x^3 + y^3 - 9xy = 0$$